Chapter 1. The Genesis of Fourier Analysis

The vibrating string and the flow of heat

Contents

1.1 The vibrating string **2**

1.1.1 Derivation of the wave equation 2

1.1.2 Solution to the wave equation 2

1.1.3 Example: the plucked string 3

1.2 The heat equation **3**

1.2.1 Derivation of the heat equation 3

1.2.2 Steady-state heat equation in the disc 4

1.1 The vibrating string

Motivation. Fourier analysis was born from two physical problems: the vibrating string and the flow of heat. Both lead to linear partial differential equations whose separated solutions are sines and cosines. The general solution is then a superposition, i.e. a Fourier series, and the central question becomes: which functions can be written this way?

Simple harmonic motion. A mass m on a spring with constant k obeys $-ky(t) = my''(t)$. With $c = \sqrt{k/m}$ the solutions are $y(t) = a\cos ct + b\sin ct = A\cos(ct-\varphi)$, with **amplitude** $A = \sqrt{a^2+b^2}$, **period** $2\pi/c$ and **frequency** $c/2\pi$; here $a = y(0)$, $b = y'(0)/c$. In complex form $e^{ix} = \cos x + i\sin x$ (Euler), $\cos x = \frac{e^{ix}+e^{-ix}}{2}$, $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$.

Standing and travelling waves. A **travelling wave** has the form $u(x,t) = F(x-ct)$ (a profile moving with speed c). A **standing wave** has the form $u(x,t) = \varphi(x)\psi(t)$: the profile keeps its shape and only its amplitude varies in time.

1.1.1 Derivation of the wave equation

Model a string of length L, density ρ and tension τ fixed at its ends by N particles of mass ρh ($h = L/N$) at $x_n = nh$, connected by springs. Let $y_n(t)$ be the vertical displacement of the n-th particle. Hooke's law for the vertical components of the tensions gives

$$\rho h\, y_n''(t) = \frac{\tau}{h}\big(y_{n+1}(t) + y_{n-1}(t) - 2y_n(t)\big).$$

As $h \to 0$ and $y_n(t) \to u(x,t)$, the right side divided by h tends to $\tau\,\partial_x^2 u$, giving the **wave equation**

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \qquad c = \sqrt{\tau/\rho}. \tag{1.1}$$

(Changing $x \mapsto x/L$, $t \mapsto ct/L$ we may take $c = 1$, $L = \pi$.) **Invariance properties:** the equation is linear and invariant under translations in x and t.

1.1.2 Solution to the wave equation

d'Alembert's solution. Introduce $\xi = x+t$, $\eta = x-t$ and $v(\xi,\eta) = u(x,t)$. The wave equation becomes $\frac{\partial^2 v}{\partial\xi\partial\eta} = 0$, so $v = F(\xi) + G(\eta)$:

$$u(x,t) = F(x+t) + G(x-t),$$

the sum of two travelling waves. On the real line with $u(x,0) = f(x)$, $\partial_t u(x,0) = g(x)$:

$$u(x,t) = \frac{f(x+t)+f(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} g(y)\,dy. \tag{1.2}$$

Check [Supplement]: $F+G = f$ and $F'-G' = g$ give $F = \frac{f}{2} + \frac{1}{2}\int_0^x g + C$, $G = \frac{f}{2} - \frac{1}{2}\int_0^x g - C$. On $[0,\pi]$ with $u(0,t) = u(\pi,t) = 0$, extend f, g to $[-\pi,\pi]$ as odd functions and then 2π-periodically; (1.2) then satisfies the boundary conditions.

Superposition of standing waves. Separating variables, $u(x,t) = \varphi(x)\psi(t)$ gives $\frac{\psi''}{\psi} = \frac{\varphi''}{\varphi} = \lambda$, a constant. Boundedness forces $\lambda = -m^2 \le 0$, so $\psi(t) = A\cos mt + B\sin mt$ and $\varphi(x) = \tilde{A}\cos mx + \tilde{B}\sin mx$; $\varphi(0) = \varphi(\pi) = 0$ forces $\varphi(x) = \sin mx$ with $m = 1, 2, \dots$. The standing waves are

$$u_m(x,t) = (A_m\cos mt + B_m\sin mt)\sin mx,$$

the m-th **harmonic** ($m=1$: **fundamental tone**; $m\geq 2$: **overtones**, of higher frequency $m/2\pi$). By linearity

$$u(x,t)=\sum_{m=1}^{\infty}(A_m\cos mt+B_m\sin mt)\sin mx \tag{1.3}$$

is a candidate solution, and the initial condition $u(x,0)=f(x)$ requires

$$f(x)=\sum_{m=1}^{\infty}A_m\sin mx. \tag{1.4}$$

Fourier's question: can every (reasonable) f on $[0,\pi]$ with $f(0)=f(\pi)=0$ be written as in (1.4)? If so, the orthogonality $\int_0^\pi \sin mx\sin nx\,dx=\frac{\pi}{2}\delta_{mn}$ forces

$$A_m=\frac{2}{\pi}\int_0^\pi f(x)\sin mx\,dx,$$

and similarly B_m is determined by $\partial_t u(x,0)=g(x)=\sum mB_m\sin mx$. Extending f oddly to $[-\pi,\pi]$, (1.4) is the special case of

$$F(x)=\sum_{m=-\infty}^{\infty}a_m e^{imx},\qquad a_m=\frac{1}{2\pi}\int_{-\pi}^{\pi}F(x)e^{-imx}\,dx. \tag{1.5}$$

1.1.3 Example: the plucked string

Pluck the string at $p\in(0,\pi)$ to height h with zero initial velocity: $f(x)=\frac{xh}{p}$ for $0\leq x\leq p$, $f(x)=\frac{h(\pi-x)}{\pi-p}$ for $p\leq x\leq\pi$, $g=0$. Then $B_m=0$ and, integrating by parts,

$$A_m=\frac{2h\sin mp}{m^2p(\pi-p)},\qquad u(x,t)=\sum_{m\geq 1}A_m\cos mt\sin mx.$$

Since $\sum|A_m|<\infty$, the series converges uniformly, and u is continuous. In general, though, (1.3) may fail to be twice differentiable, so it solves the wave equation only in a generalised sense (compare d'Alembert's formula).

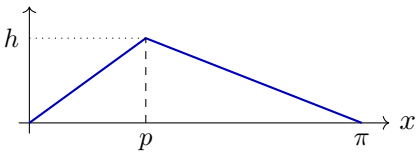


Figure. *The plucked string at $t=0$.*

1.2 The heat equation

Motivation. Heat flow leads to the heat and Laplace equations. Steady-state temperatures in a disc are again obtained by separating variables, this time in polar coordinates, and lead to Fourier series in the angle θ.

1.2.1 Derivation of the heat equation

For a plate $\mathbb{R}^2$ with temperature $u(x,y,t)$, specific heat σ and thermal conductivity κ: the heat in a small square S is $\sigma\iint_S u$, and the heat flux across ∂S is $\kappa\int_{\partial S}\partial u/\partial n$ (Fourier's law: heat flows from

hotter to colder, proportionally to the gradient). Equating the rate of change of heat in S with the flux, dividing by the area and shrinking S:

$$\frac{\partial u}{\partial t} = \frac{\kappa}{\sigma}\Big(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\Big) = \frac{\kappa}{\sigma}\,\triangle u. \tag{1.6}$$

In a **steady state** $\partial u/\partial t = 0$, and u satisfies the **Laplace equation** $\triangle u = 0$; its solutions are **harmonic functions**.

1.2.2 Steady-state heat equation in the disc

Dirichlet problem. Find u with $\triangle u = 0$ in the unit disc $D = \{x^2 + y^2 < 1\}$ and $u = f$ on the circle C. In polar coordinates

$$\triangle u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}.$$

Separating $u = F(r)G(\theta)$, $\frac{r^2F''+rF'}{F} = -\frac{G''}{G} = \lambda$. G is 2π-periodic, so $\lambda = m^2$ and $G(\theta) = e^{\pm im\theta}$, $m \geq 0$. Then $r^2F'' + rF' - m^2F = 0$ has solutions $F = r^{\pm m}$ $(m \geq 1)$, $F = 1, \log r$ $(m = 0)$; boundedness at 0 leaves $F = r^{|m|}$. So the candidate solution is

$$u(r,\theta) = \sum_{m=-\infty}^{\infty} a_m r^{|m|} e^{im\theta}, \qquad f(\theta) = \sum a_m e^{im\theta}. \tag{1.7}$$

Substituting $a_m = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\varphi)e^{-im\varphi}d\varphi$ and summing the geometric series formally,

$$u(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta-\varphi)f(\varphi)\,d\varphi, \qquad P_r(\gamma) = \sum_{m\in\mathbb{Z}} r^{|m|}e^{im\gamma} = \frac{1-r^2}{1-2r\cos\gamma + r^2},$$

the **Poisson kernel**; this is made rigorous in Chapter 2.